

Risk-aware Path Planning via Probabilistic Fusion of Traversability Prediction for Planetary Rovers on Heterogeneous Terrains

- The old approach $\rightarrow$ Scientists tend to use "Ground-in-the-loop (GITL)" technique in which data, after a whole Martian day, reaches to team of scientists on Earth where they decide the robot has to travel where

From this approach $\rightarrow$ 28.9 metres per sol day



↓
• New method of CNN classifier is used → The approach is good for homogeneous surface

→ In practicality, the surface of MARS is heterogeneous

→ cause of this method, Rover was not able to use controller system

properly cause of no idea of slope [it is safe or not]

→ as a result, many missions reported the sinking, non-traversal

of terrain

→ as a result, many researchers reported the striking, non-traversability of drone

- Not a new, but a method of using Physics baseline models (basically Gaussian Process is proposed).

→ This ~~type~~ time, thanks to sensor connected with wheels and evolution in terrain domain,

→ The Rover was able to identify safe or unsafe slope by Regression-based technique of Rover-slope safe relationship. —:

{ The model trains itself on the basis of surface (whether sandy or bedrock or any other) and identifies the safe slope to climb or descend. }

→ This method not only provides this but also "~~uncertainty~~ epistemic Uncertainty by providing Uncertainty Quantification

probabilistically models the traversability (i.e., slips) at an

edge eee for a terrain class C as

$$p(s_e | \phi_e, \theta) = \mathcal{N}(\mu_e(\phi_e), \sigma_e^2(\phi_e))$$

most likely slip value = predictive mean

taking from already trained dataset

predictive variance

- U-net Architecture Decoder $\rightarrow$ Takes the data as same as other CNNs
 $\rightarrow$ maxpooling is applied and same data is then decoded using softmax function

$\rightarrow$ Giving probabilistic values of observed thing
in pixels
Very necessary cause we
don't know new object, surface or
sometimes lightning.

sometimes lightning.

- Now both U-net Architecture and GP are combined to solve each other's problem

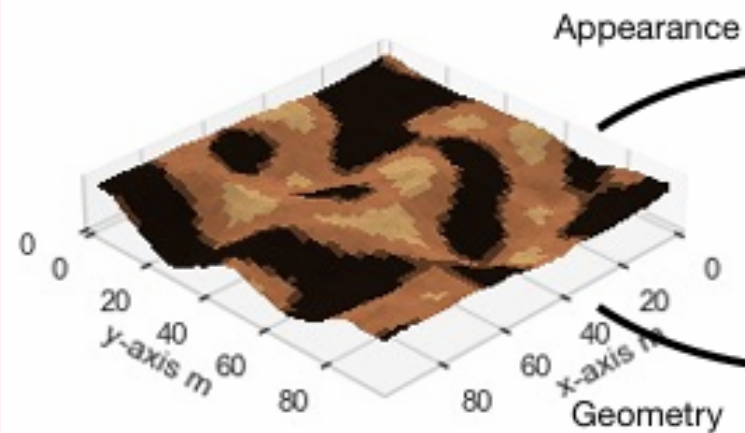
Problem of U-net :- Cause of many uncertain things, like lightning or new kind of object, classifier can sometimes do mistake, resulting in a false slip value.

Problem of GP :- It assumes the whole surface to be homogeneous, meaning suppose we have 15° slope bedrock and 15° slope sandy terrain and suppose rover has trained on both datasets differently whether it is unsafe or not, cause of this when a new slope comes of either bedrock or sandy, the rover does not know (cause we fed slope in the formula) difference in the class, so basically it will try to average for both data

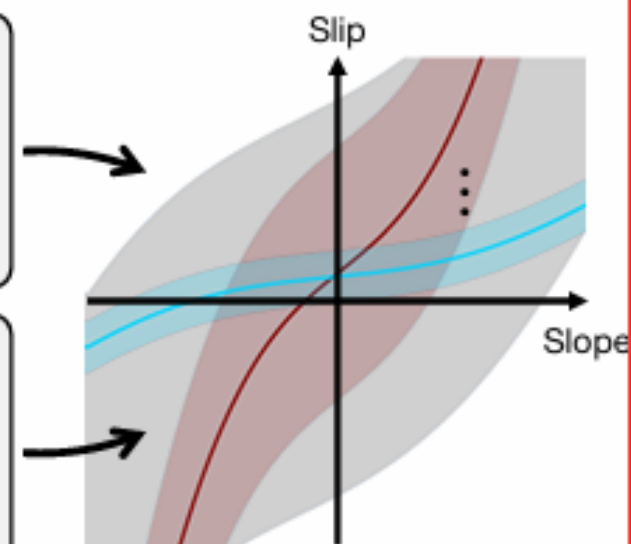
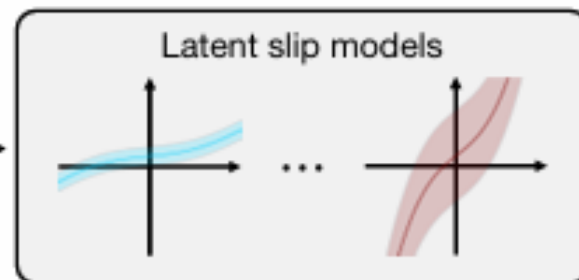
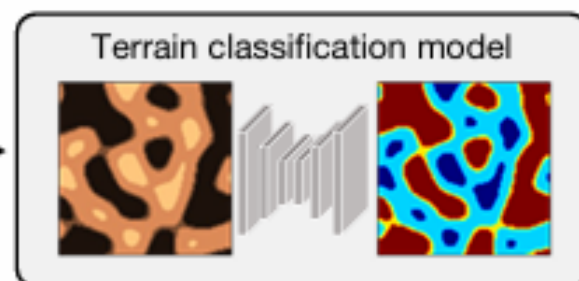
↑
so basically we need a classifier which will update the class of terrain or input in real time for new environments
which is what this paper is about.

- formula for probabilistic fusion:- $P_{e:v \rightarrow v}(s) = \sum_{c \in C} \underbrace{P_v(c)}_{\text{U-net}} \times \underbrace{P_e(s|c)}_{\text{GP}}$
probability of terrain at vertex v belongs to class c
- Now, what belongs to

Problem Instance



Probabilistic Fusion of ML Models



(a) ML prediction models → (b) Probabilistic fusion via MGP

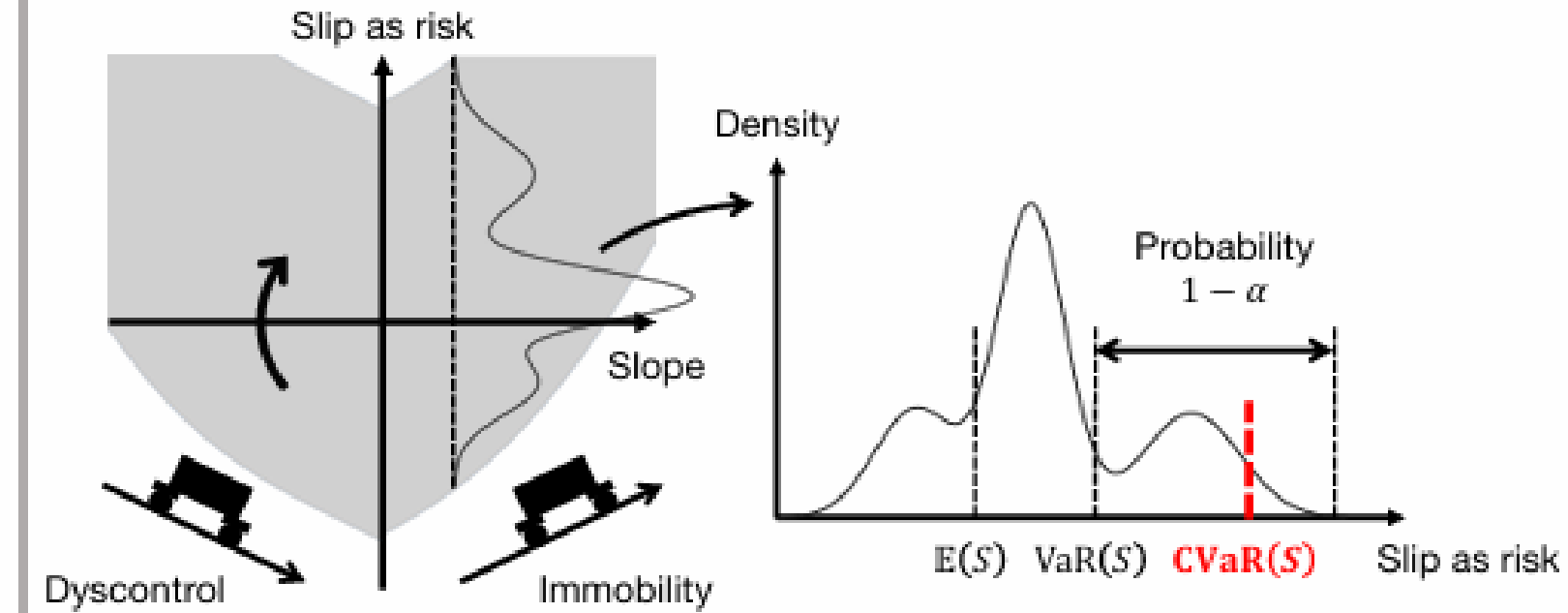
- U-net GIP
- Now, what belongs to our CCA usecase $\rightarrow$ how to reduce the cost function



basically we can take some risks to search to the preferred location
while considering that some slips or paths we definitely had to avoid.

- This is where we use CVAR (Conditional Value at Risk) $\rightarrow$ Done by Monte Carlo sampling.
 - $\rightarrow$ In this basically, we check the slip where the $CVAR > VAR$
 - $\rightarrow$ Through this we can penalize the paths where catastrophic outcomes are expected.
- $\rightarrow$ VAR is just the ~~average of~~ max. output which we are expecting if say slip
... .. tbn this part is not that much clear

Risk-aware Cost Estimation



→ (c) Slip as risk distribution → (d) Risk evaluation for slip

- At last, we compute a cost function and give it to A* algorithm, so A* finds us a path planned with the least cost function.

$$f_{\text{cost}}(e: v \rightarrow v') = \frac{\|x(v) - x(v')\|}{U_{\text{risk}}(e)}$$

$x(v)$ denotes the 3D posⁿ at a vertex v .
 $U_{\text{risk}}(e)$ is the risk of edge e .
 basically $\frac{\text{actual velocity}}{\text{circumferential velocity}}$

rover velocity when ~~transferring~~ ~~an edge~~
 accounting for comprehensive risks

• Implementation time :- Datasets used:- (i) Std (ii) AA (iii) Errorneous.



Result: model performed well in every dataset case &
with max. solved rate and success rate and min.



slip experienced.

but obviously time was taken more as compared to other
(cause of CVAR) methods

TABLE I
QUANTITATIVE PATH PLANNING RESULTS OVER DIFFERENT DATASETS

Standard (Std) Dataset					
		Sol.	Succ.	T_{total}	s_{max}
SGP [9]	+EV	100	71	22.6 ± 3.7	58.3 ± 47.0
	+VaR	100	74	22.6 ± 3.9	55.6 ± 45.4
	+CVaR	100	75	22.6 ± 3.9	55.3 ± 45.8
MGP	+EV	100	75	22.5 ± 3.5	56.1 ± 44.4
	+VaR	100	93	22.9 ± 4.4	47.5 ± 30.7
Ours (MGP+CVaR)		100	96	23.2 ± 5.2	45.9 ± 24.2
Erroneous Slip (ES) Dataset					
		Sol.	Succ.	T_{total}	s_{max}
SGP [9]	+EV	100	60	23.0 ± 3.5	70.3 ± 40.7
	+VaR	98	65	24.3 ± 5.1	68.8 ± 44.7
	+CVaR	93	62	25.0 ± 5.7	67.9 ± 44.8
MGP	+EV	100	71	23.1 ± 4.4	68.2 ± 38.2
	+VaR	98	77	25.2 ± 12.7	63.1 ± 38.8
Ours (MGP+CVaR)		90	77	27.2 ± 37.4	60.5 ± 37.1
Ambiguous Appearance (AA) Dataset					
		Sol.	Succ.	T_{total}	s_{max}
SGP [9]	+EV	100	11	22.1 ± 1.7	92.9 ± 27.4
	+VaR	100	7	24.8 ± 4.0	95.8 ± 19.3
	+CVaR	100	2	26.7 ± 1.8	96.3 ± 19.3
MGP	+EV	100	38	24.0 ± 5.6	81.4 ± 32.9
	+VaR	98	95	25.7 ± 5.3	64.5 ± 22.3
Ours (MGP+CVaR)		96	95	26.4 ± 6.3	63.7 ± 21.6

